THE CRITICAL SPECIFIC HEAT OF GADOLINIUM

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Results are presented of a comprehensive analysis of previously reported measurements of the critical thermal expansion of gadolinium which suggest that the dominant singularity at T_c in the specific heat is a discontinuity. New measurements of the critical specific heat indicate a higher T_c and a sharper transition at T_c than previously reported, and generally support the conclusions of the expansion experiments.

It has been suggested recently from measurements of the thermal expansion of pure single crystal samples, that gadolinium does not behave as a simple Heisenberg ferromagnet in that the dominant singularity at T_c in the thermal expansion, and presumably therefore in the specific heat as well, appears to be a discontinuity [1]. We report here further evidence for this suggestion including results for the specific heat of one of the samples used in the thermal expansion measurements, and details of a more thorough analysis of the expansion data.

The specific heat of a crystal of mass 1.24 g and a mean resistance ratio $[\rho(293)/\rho(4.2)]$ of 165 was measured between 200–320 K in a conventional automatic continuous warming calorimeter at the Australian National University [6]. The accuracy of the calorimeter had previously been assessed with a 40 g copper sample to be within 0.2%. With the much smaller gadolinium sample the scatter in the data was as much as 2%, although, except very near T_c, this was offset by the high density of data that was recorded. Three experiments were performed at two different warming rates, the results generally agreeing to within 1%, although there is evidence of a slight dependence on warming rate very near T_c. As the data density is higher for the slow warming experiment only these results have been fully evaluated. The results are shown in fig. 1 with, for comparison, results obtained with a large polycrystalline sample in the same apparatus [2] and the results of Griffel et al. [3]. The agreement between all sets of data is very good below T_c although above T_c there is a difference between the results of Griffel et al. and those obtained with our apparatus. The specific heat of the single crystal sample has a maximum value of 66 J mol^{-1} K^{-1} at $T_{MAX} = 293.55$ K. T_{MAX} is higher than previously reported though it is in good agreement with the corresponding value for the thermal expansion. Rounding effects near T_c are considerably less than previously observed as expected for a sample of high purity.

The specific heat and the thermal expansion data have been fitted to the following equations:

$$y = (A^{\pm}/\alpha)|\epsilon|^{-\alpha} + B + C\epsilon, \quad (1)$$

$$y = (A^{\pm}/\alpha^{\pm})|\epsilon|^{-\alpha^{\pm}} + B + C\epsilon, \quad (2)$$

$$y = (A^{\pm}/\alpha)|\epsilon|^{-\alpha} + B^{\pm} + C\epsilon. \quad (3)$$

where $\epsilon = (T - T_c)/T_c$, and $+$, $-$ refer to $T > T_c$ and $T < T_c$ respectively. Eq. (1) is the form expected from theory for the specific heat of a simple ($n = 3$, $d = 3$) Heisenberg ferromagnet with $\alpha = -0.125$ and $A^+/A^- = 1.4$. It follows that ideally the specific heat should be continuous through T_c and equal to B at T_c [4]. Eqs. (2) and (3) are essentially empirical, each having one more fitting parameter than (1). Eq. (2) corresponds to a breakdown of scaling ($\alpha^+ \neq \alpha^-$) whilst for $\alpha < 0$, eq. (3) corresponds to a discontinuity at T_c($B^+ \neq B^-$). In fitting the data T_c and α were stepped over a range of values and for each combination of T_c and α a least squares fitting program was used to find the other parameters. The quality of fitting was judged initially by an error parameter E equal to a weighed sum of squares of deviations of the data from the fitted equation. Further checks of the goodness of fit were made by plotting the deviations.

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TABLE 1

A summary of the fits of α_C and C_p for Gd to eqs. (1), (2) and (3). (The units of A, B, C are (J mol^{-1} K^{-1}) for C_p and ($\times 10^{-6}$ K^{-1}) for α_c)

Parameter	Thermal expansion α_C			Specific heat C_p		
	Eq. (1)	Eq. (2)	Eq. (3)	Eq. (1)	Eq. (2)	Eq. (3)
T_c(K)	293.54	293.54	293.60	293.54	293.54	293.60
α^+	} − 0.02	−0.12	} − 0.32	} − 0.02	−0.09	} − 0.30
α^-		−0.39			−0.42	
A^+	−8.7	−16.3	−36.5	3.2	3.8	11.4
A^-	−5.2	−52	−36.8			16.5
B^+	} − 358	}98	−49	}186	}68	49
B^-			−107			71
C	−342	−368	−57	89	−28	0
E	980	760	13	29	27	25

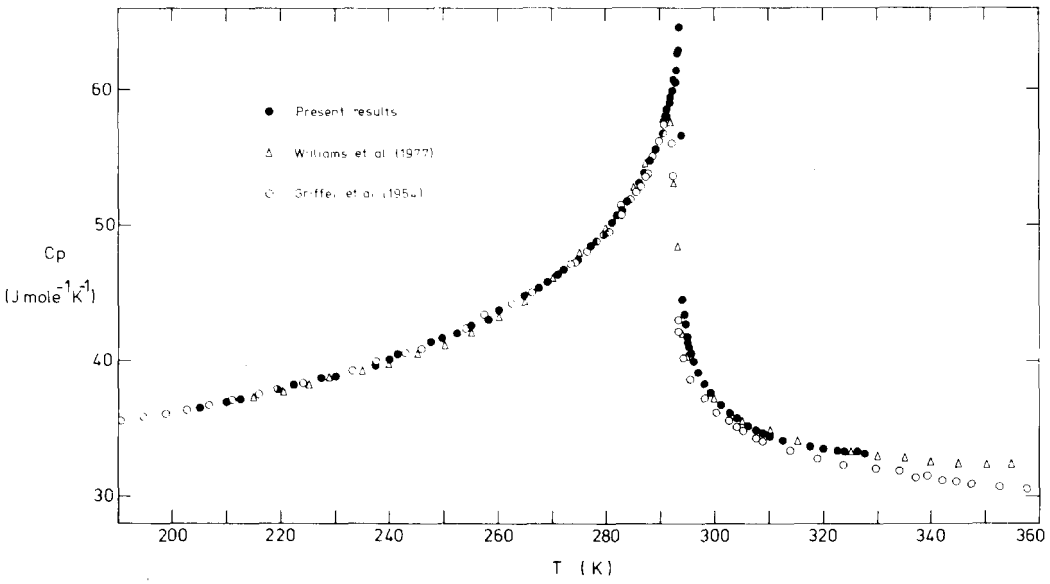


Fig. 1. The specific heat of gadolinium as a function of temperature.

Due to thermal expansion the effective value of T_c for Gd is temperature dependent and the ϵ value for a given temperature should be calculated using the effective value of T_c for that temperature. This correction has been considered for both the specific heat and thermal expansion data but makes no significant difference to T_c or α or the quality of the fits, although small changes occur in the other parameters. The correction from C_p to C_v has also been considered but has no significant effect. Table 1 contains a summary of the fits for the specific heat C_p and c-axis expansion α_C using uncorrected ϵ values in the range $8 \times 10^{-4} < \epsilon < 2 \times 10^{-2}$.

The only clear minimum in E combined with a satisfactory fit occurs for the expansion data fitted to eq. (3). The other fits given in table 1 are optimum for the values of T_c specified, but can be improved by allowing T_c to fall well below T_{MAX}. This is particularly marked in fitting α_C to eq. (1) which may account for the very low value of T_c found by Dolejsi in fitting his data to the same equation [5]. However it is considered unlikely that T_c should be below T_{MAX} and in any case the fits

for low T_c values judged from plots of the deviations are not good and the parameters do not agree with theory.

The evidence of the thermal expansion clearly suggests that a discontinuity is the dominant singularity at T_c as previously reported [1]. Due to the smaller singularity in C_p and the relatively larger scatter in the data the analysis of the specific heat is much less conclusive, the goodness of fit parameter E hardly varying over a wide range of fitting parameters. However examination of the deviations of experimental points from the fitted curves shows that the fit to eq. (2) is considerably better than the fit to eq. (1) in agreement with the thermal expansion. We are presently making further measurements of the specific heat of gadolinium in order to improve the accuracy, especially very near T_c.

References

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